Chidubem (Chidu) Umeh

Orlando, FL | U.S. Citizen, eligible for security clearance | 617-750-4536 | Chidumeh@gmail.com | chidu.dev | linkedin.com/in/chidubem-umeh | github.com/Chidubem5

Summary

Physicist and quantitative analyst targeting operations research, modeling and simulation, and data science roles in defense and federal settings. Builds statistical models, simulations, and production data pipelines end to end, with a focus on finding where models diverge from observed behavior.

Skills

Languages: Python (Pandas, NumPy, Scikit-learn), SQL (PostgreSQL), JavaScript, R, Fortran
Methods: Statistical modeling, computational simulation, quantitative risk modeling, data pipeline engineering, data visualization
Tools: Tableau, Excel, Git, Jupyter, VASP, LLM assisted development (Claude Code, OpenAI Codex)

Experience

University of Central Florida, Orlando, FL | Graduate Research and Teaching Assistant **Jan 2024 to Aug 2026**

• Ran 50+ density functional theory simulations in VASP, modifying Fortran source code, to model defects in hexagonal boron nitride.
• Built Python pipelines to validate spectrometer data from a 9 T cryostat, flagging where model predictions diverged from measurements.
• Taught and assessed 250+ undergraduate physics students, turning complex concepts into clear, practical explanations.

Autura | Data Analyst Intern **Apr 2023 to Aug 2023**

• Joined U.S. Census demographics with internal logistics records in SQL, producing analytics ready datasets for executive reporting.
• Built Tableau dashboards and briefed leadership on findings that informed market expansion decisions.

Massachusetts Financial Services (MFS), Boston, MA | Information Technology Intern **Jun 2021 to Aug 2021**

• Delivered a production solution on an enterprise low code platform that extracted data from unstructured PDFs at 90% accuracy.

Bates College, Lewiston, ME | Research Data Analyst **Mar 2019 to May 2022**

• Analyzed telescope survey data on starburst galaxies, coauthoring a peer reviewed publication and presenting at the 235th AAS meeting.

Projects and Awards

U.S. Gun Violence Intelligence Tool | Python **Jan 2026 to Jun 2026**

• Built an end to end pipeline turning 300,000+ CDC WONDER records across 25 years into a deployed tool for disparity analysis.

Erdős Institute Top Project Award | Quantitative Finance Bootcamp **Nov 2025**

• Built a Black-Scholes-Merton volatility model over 1,000+ trading days, finding a 0.27 implied vs. realized correlation; top project of 52.

Splittank | Founder and Developer | JavaScript **Jan 2026 to Present**

• Shipped a full stack cost splitting app integrating live EIA and AAA fuel prices, toll data, and four payment platforms.

Education

University of Central Florida | M.S. Physics, Concentration in Spintronics **May 2026**

Bates College | B.S. Physics, Concentrations in Applying Mathematical Methods and Digital and Computational Studies **May 2022**